

Immigration and Credit in America

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TransUnion (the data provider) has the right to review the research before dissemination to ensure it accurately describes TransUnion data, does not disclose confidential information, and does not contain material it deems to be misleading or false regarding TransUnion, TransUnion's partners, affiliates or customer base, or the consumer lending industry.

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- Immigrants start life in the U.S. *without* a U.S. credit report
 - Any foreign credit history generally unobserved to U.S. lenders

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

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- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
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What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, (with fixed effects for birth year & ZIP5 geography)

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- U.S. consumer credit reporting data matched with immigration data

What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
 - Immigrants vs. same-age Non-Immigrants & Immigrants arriving at different ages

Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)
- **Delayed entry** into credit system & **lower long-term credit access**

Timing of immigration best explains our results (“history dependence”)

Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Azoulay et al., 2022; Bernstein et al., 2025)
- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1985; Abramitzsky & Boustan, 2017)
- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022; Bailey et al., 2022)
- We contribute to the household finance literature that has largely not studied immigrants (e.g., Dobbie et al., 2021; Zillessen, 2022; Bakker et al., 2025; Gorback & Schubert, 2025)

Roadmap For Presentation

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

Data & Institutional Details

University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang
(2025, *Journal of Economic Literature*)

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American Community Survey (ACS), 2010

- Validate our immigration classification

Survey of Consumer Finances (SCF), 2022

- Evaluate mechanisms behind our results

Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5 digits of the 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

- validated in prior work (Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)
- **“Immigrants”**: assigned SSN at age 21+. **“Non-Immigrants”**: assigned SSN at age < 21

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Classifying Immigrants In The BTCCP

1. Create lookup table mapping first 5 digits of SSN to SSN assignment year (*SSN Year*)
2. TransUnion merged to their data & returned SSN Year (no SSNs) for each consumer
3. We calculate age of SSN assignment: $SSN\ Age = SSN\ Year - Birth\ Year$

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Entrant Sample Of Birth Years 1975 to 1987 With SSN Ages 0 to 29:

6,122,932 consumers – 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

- Immigrant credit access may be adversely affected by **information frictions**:
 1. **Not having a credit report on arrival**
 - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
 2. **After entering the credit system, have a shorter history than Non-Immigrants**
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- Different cultural **tastes** & informational barriers for consumers
 - Differences in familiarity & willingness to use debt, understanding of how to build credit score
- Lenders use credit scores (+ other information) in decisions
 - Can use immigration status, can't use protected characteristics (e.g., national origin, race) under Equal Credit Opportunity Act (ECOA)

Credit Market Entry

Main Cross-Sectional Regression Specifications

β_1 : average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (1)$$

- Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

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β_2 : marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

- Cluster standard errors by birth year

Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

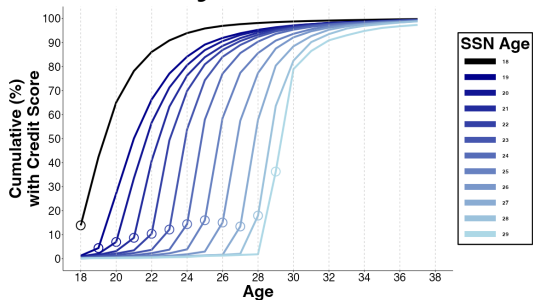
	Age at First Credit Report	
	(1)	(2)
Immigrant	4.98*** (0.14)	1.81*** (0.09)
SSN Age		0.74*** (0.01)
F.E. Birth Year	X	X
F.E. First Zip5	X	X
R^2	0.264	0.289
Mean, SSN Age <21	19.86	19.86

$N = 6,122,932$, *** 0.5%, **, 1%, * 5% statistical significance

Creditworthiness

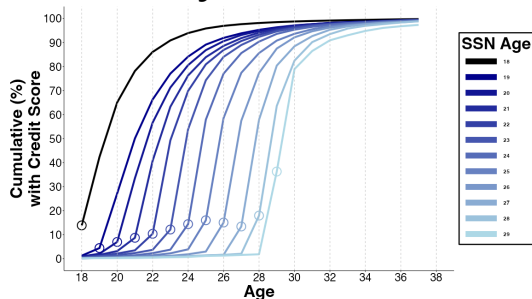
Once Scored, Immigrants Have Higher Credit Scores Than Non-Immigrants

A. Any Credit Score

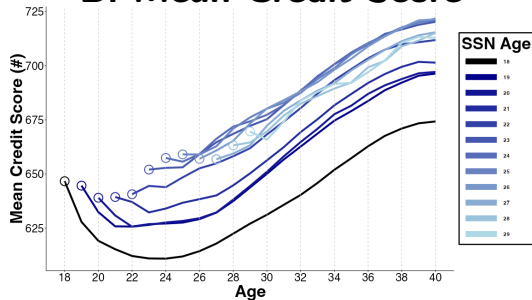


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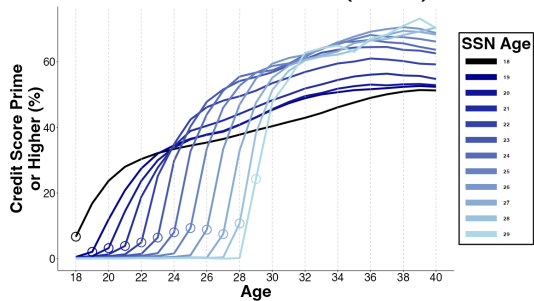


B. Mean Credit Score



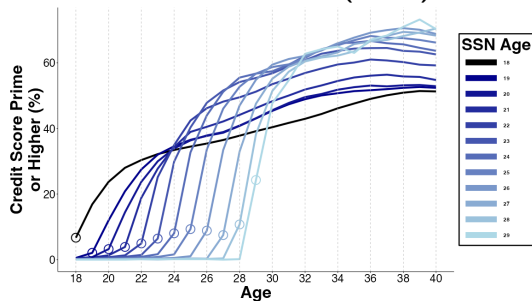
Immigrants (vs. Non-Immigrants) +27 points average scores by age 30, +33 by age 40

C. Prime or Higher Credit Score (660+)



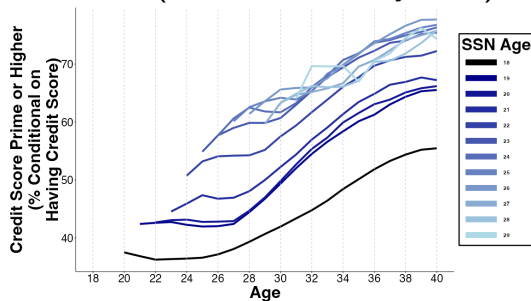
Positive Selection Of Immigrants In Credit Score Distribution

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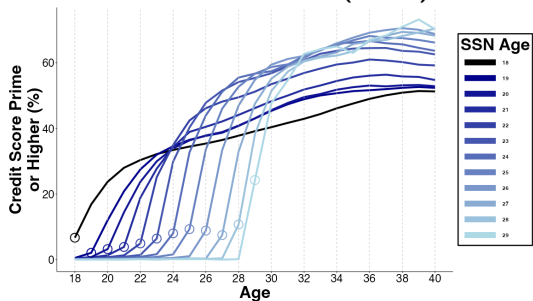
Immigrants: $\downarrow$ Pr(subprime), $\uparrow$ Pr(superprime)

D. Prime or Higher Credit Score (Conditional On Any Score)

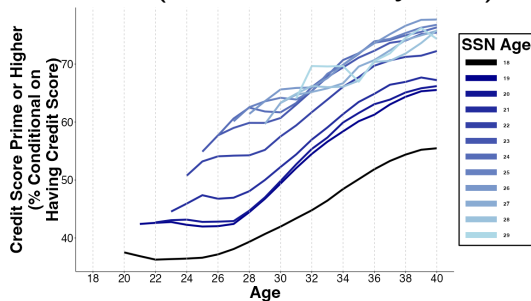


Positive Selection Of Immigrants In Credit Score Distribution

C. Prime or Higher Credit Score (660+)



D. Prime or Higher Credit Score (Conditional On Any Score)



Immigrants: $\downarrow$ $\Pr(\text{subprime})$, $\uparrow$ $\Pr(\text{superprime})$

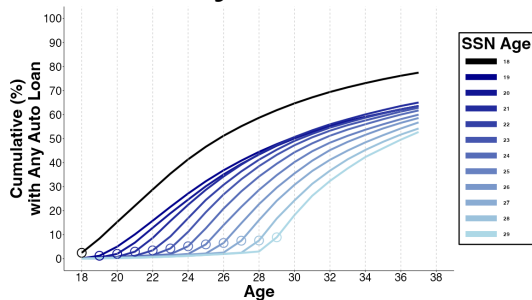
More creditworthy behaviors: $\downarrow$ delinquency rates & $\downarrow$ credit card utilization

$\downarrow$ 5% delinquency rates conditional on credit score

Credit Access

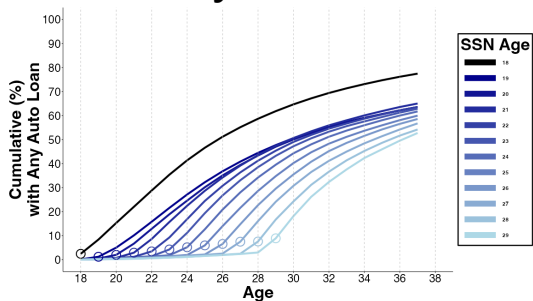
Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

A. Any Auto Loan

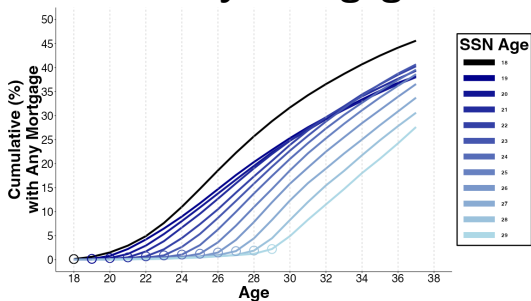


Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

A. Any Auto Loan



B. Any Mortgage



By Age 37, Immigrants 17% (16%) Less Likely To Have Auto Loan (Mortgage)

- ↓ Access For Later In Life Immigrants

Dep. Var: By Age 37, has ...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-12.99*** (0.80)	-6.87*** (0.46)	-7.18*** (0.74)	0.39 (0.53)	0.31 (0.22)	0.15 (0.16)
SSN Age		-1.44*** (0.10)		-1.77*** (0.09)		0.04 (0.03)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.039	0.039	0.062	0.063	0.025	0.025
Mean, SSN Age <21	77.22	77.22	45.46	45.46	94.93	94.93

$N = 6,122,932$, *** 0.5%, **, 1%, * 5% statistical significance

“Paired Cohorts”: Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- ‘Focal Cohort’: $SSN\ Age_c$, 22-30
- ‘Control Cohort’: $SSN\ Age_c - 1$, same birth year as focal. 68 pairs (p)
- 2,176 cohort-by-year observations ($k(p), t$), s.e. clustered by birth year

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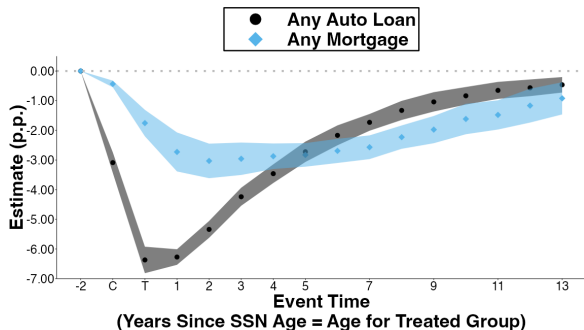
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- **Results:**

Any Auto Loan after 10 years:

−0.46 p.p., −0.42 cumulative years

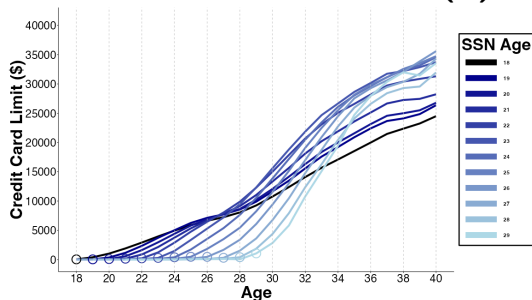
Any Mortgage after 10 years:

−0.92 p.p., −0.29 cumulative years



Immigrants' Credit Card Limits Are Initially Lower Than Non-Immigrants, But Become Higher Than Non-Immigrants Over Life Cycle

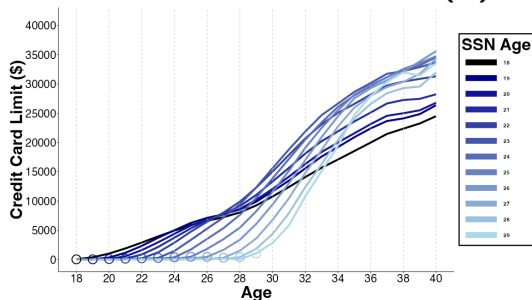
A. Credit Card Limits (\$)



Similar patterns for number of credit cards,
credit card limit per card

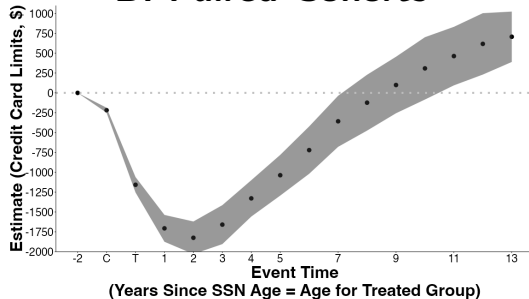
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B. Paired Cohorts



SSN Age +2 Years:

−\$1,825 (s.e. \$105)

SSN Age +13 Years:

+\$707 (s.e. \$162)

Mechanisms

What Mechanisms Explain?

- Less Credit For More Creditworthy Immigrants
- Less Credit For Later In Life Immigrants

Creditworthiness

Emigration

Credit Demand & Tastes

History Dependence

- Credit History Dependence: Financial friction of no U.S. credit report & shorter history
- Non-Credit History Dependence: less time in U.S. means...
 - fewer years of U.S. earnings
 - less far along in American life cycle

Immigrants Are Averse To Purchasing Cars With Debt (SCF, 2022)

-19% Immigrants Vs. Non-Immigrants

Dep Var:	<u>Auto Debt Attitude</u>		<u>Any Auto Debt</u>		<u>Any Auto Debt</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.41*** (4.58)	-23.02*** (7.25)	-4.27 (5.08)	-11.14 (8.27)	-0.21 (5.01)	-5.10 (7.87)
Immigration Age		0.98 (0.62)		0.88 (0.76)		0.63 (0.69)
Auto Debt Attitude					26.34*** (3.23)	26.24*** (3.24)
Demographic Controls	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.063	0.106	0.106
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14

$N = 1,813$, *** 1%, ** 5%, * 10% statistical significance

Non-Portability Of Credit Histories Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions To Increase Credit Access:

American Express Global Card Relationship

Global backing for Card Members,
wherever you move

Global Card Relationship, our Card Member-only service, makes it seamless for Card Members to apply for a new American Express® Card in select countries.

Nova Credit

International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

Equifax Canada Global Consumer Credit File

Global Consumer Credit File

Helping you grow and retain your clients
who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

Conclusions

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection & assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants
- exploit variation in age of immigration in credit data matched with immigration information

Immigrants vs. same age Non-Immigrants & Immigrants arriving at older ages:

- **Positively selected:** more creditworthy (once scored)
- **Lower long-term access** to auto loans and mortgages
 - Mortgage gap explained by timing of immigration
 - Auto loan gap partially due to auto loan-specific debt aversion
- *Timing* of immigration best explains our results (“**history dependence**”)

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Thank You!

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Entrant Sample In BTCCP

Sample Restrictions

- Birth years 1975 - 1987
- SSN Ages 0 to 29 (21 to 29 “Immigrants”)
- Credit report by 2011

Entrant Sample

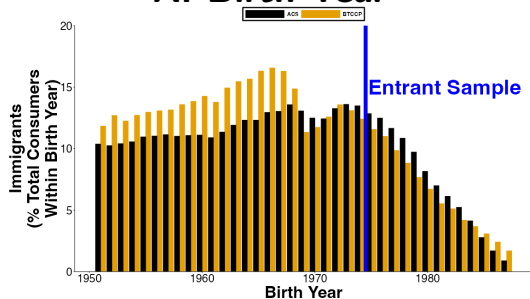
6,122,932 consumers

– 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)

Follow credit outcomes in BTCCP for ages 18 to 40

Distribution of Immigrants by...

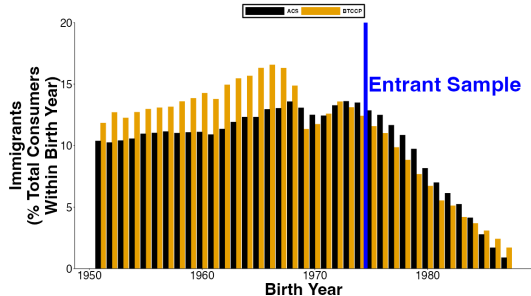
A. Birth Year



Immigrant Classification Aligns With American Community Survey (ACS) 🚀

Distribution of Immigrants by...

A. Birth Year



B. Immigration Age



BTCCP: 11.27% SSN Age 21+

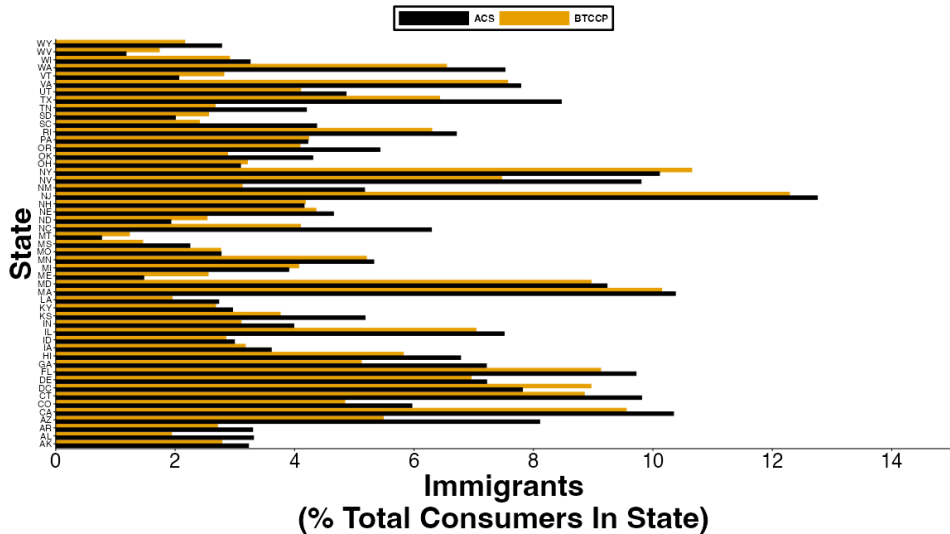
ACS: 10.2% Immigrated at Age 21+

Distribution of immigrants aligns by geography. Immigrants more mobile than non-immigrants.

“Entrant Sample”: Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry At Ages 21 To 29 & Credit Outcomes For Ages 18 To 40 🍷

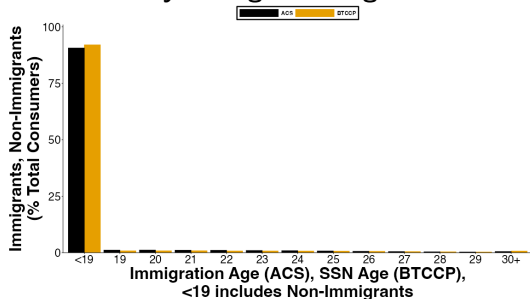
TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
 Clean Sample	 18,572,654
... SSN Age <21	16,478,940
... SSN Age 21+	2,093,714 (11.27%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
 Entrant Sample (SSN Age < 30)	 6,122,932
... SSN Age < 21	5,778,671
... SSN Age 21-29	344,261 (5.62%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

Geographic Comparison of Immigrants: ACS vs. BTCCP 🚀

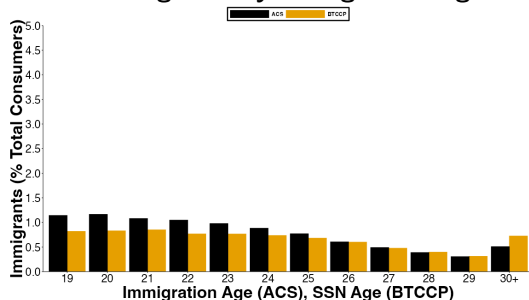


Immigrant Classification in Entrants Sample: ACS vs. BTCCP

A. Non-Immigrants and Immigrants by Immigration Age



B. Immigrants by Immigration Age

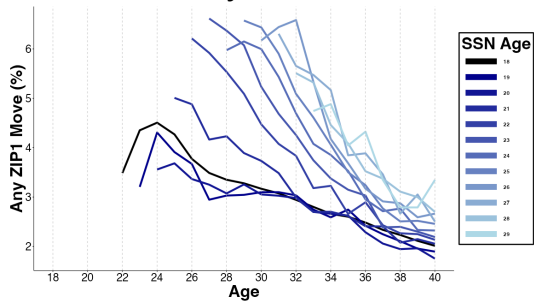


BTCCP: 5.62% SSN Age 21 to 29

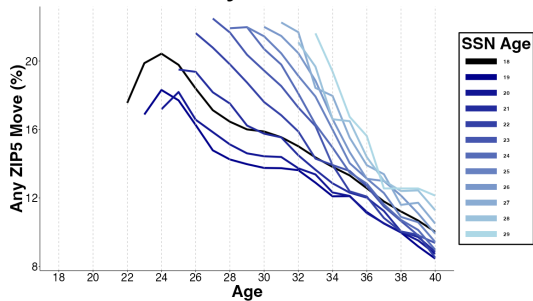
ACS: 6.57% Immigrated at Age 21 to 29

Geographic Mobility 🚀

A. Any ZIP1 Move

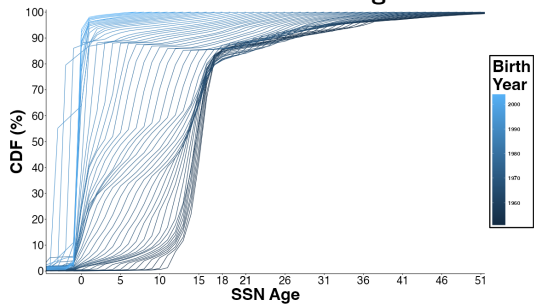


B. Any ZIP5 Move

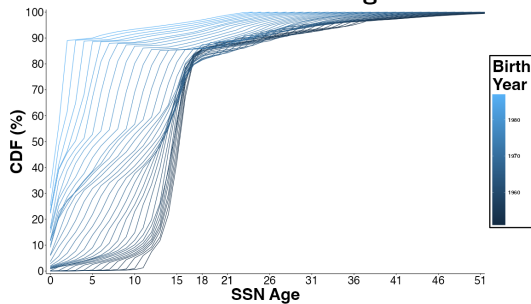


SSN Ages By Birth Year

A. Before Cleaning



B. After Cleaning



Consumers By SSN Age

SSN Age	Clean Sample	Entrant Sample
<19	16,155,157	5,677,001
19	175,194	50,475
20	148,589	51,195
21	148,019	52,450
22	141,492	47,302
23	145,218	47,214
24	149,000	45,314
25	147,899	42,005
26	142,319	36,976
27	133,147	29,383
28	123,338	24,415
29	113,435	19,202
30+	849,847	44,642

Sample Frame and Timing of SSN Assignment

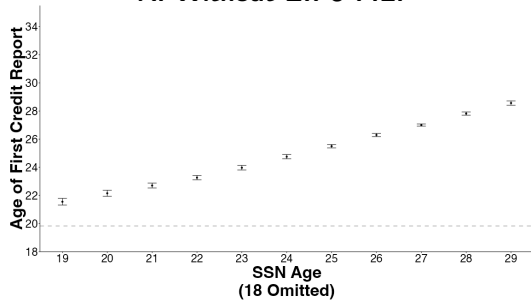
Calendar year	Birth year													
	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	
1993	≤18													
1994	19	≤18												
1995	20	19												
1996	21	20	≤18											
1997	22	21	20	≤18										
1998	23	22	21	20	19	≤18								
1999	24	23	22	21	20	19	≤18							
2000	25	24	23	22	21	20	19	≤18						
2001	26	25	24	23	22	21	20	19	≤18					
2002	27	26	25	24	23	22	21	20	19	≤18				
2003	28	27	26	25	24	23	22	21	20	19	≤18			
2004	29	28	27	26	25	24	23	22	21	20	19	≤18		
2005	30	29	28	27	26	25	24	23	22	21	20	19	≤18	
2006	31	30	29	28	27	26	25	24	23	22	21	20	19	
2007	32	31	30	29	28	27	26	25	24	23	22	21	20	
2008	33	32	31	30	29	28	27	26	25	24	23	22	21	Age at SSN assignment
2009	34	33	32	31	30	29	28	27	26	25	24	23	22	
2010	35	34	33	32	31	30	29	28	27	26	25	24	23	
2011	36	35	34	33	32	31	30	29	28	27	26	25	24	
2012	37	36	35	34	33	32	31	30	29	28	27	26	25	
2013	38	37	36	35	34	33	32	31	30	29	28	27	26	
2014	39	38	37	36	35	34	33	32	31	30	29	28	27	
2015	40	39	38	37	36	35	34	33	32	31	30	29	28	
2016	41	40	39	38	37	36	35	34	33	32	31	30	29	
2017	42	41	40	39	38	37	36	35	34	33	32	31	30	
2018	43	42	41	40	39	38	37	36	35	34	33	32	31	
2019	44	43	42	41	40	39	38	37	36	35	34	33	32	
2020	45	44	43	42	41	40	39	38	37	36	35	34	33	Years of data in our sample
2021	46	45	44	43	42	41	40	39	38	37	36	35	34	
2022	47	46	45	44	43	42	41	40	39	38	37	36	35	
2023	48	47	46	45	44	43	42	41	40	39	38	37	36	
2024	49	48	47	46	45	44	43	42	41	40	39	38	37	

Summary Statistics

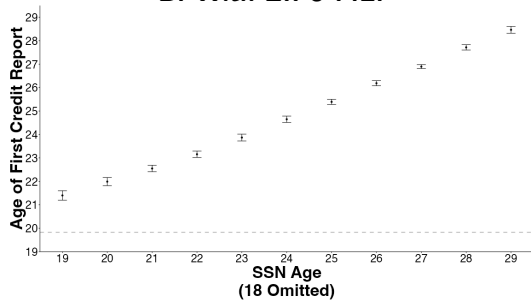
	<u>Non-Immigrants</u> <i>(SSN Age $\geq$ 21)</i>		<u>Immigrants</u> <i>(SSN Age 21 to 29)</i>	
	Mean	<i>N</i>	Mean	<i>N</i>
Credit Report				
Age at First Credit Report	19.86	5,778,671	25.21	344,261
Credit Score				
Any Score by Age 30 (%)	97.57	5,778,671	88.14	344,261
Credit Score at Age 30	626.1	5,461,785	660.7	293,349
Any Score by Age 40 (%)	99.68	4,487,804	99.14	312,391
Credit Score at Age 40	659.5	4,179,752	698.9	270,177
Auto Loan				
Any Auto Loan (%)	79.97	5,778,671	66.07	344,261
Age at First Auto Loan	27.62	4,621,146	31.61	227,439
Mortgage				
Any Mortgage (%)	50.06	5,778,671	46.41	344,261
Age at First Mortgage	30.59	2,892,536	33.79	159,758
Credit Card				
Any Credit Card (%)	94.51	5,778,671	94.00	344,261
Age at First Credit Card	24.42	5,461,356	27.97	323,607
Credit Card Limits at Age 30	\$11,733	5,778,671	\$11,193	344,261
Credit Card Limits at Age 40	\$22,477	4,487,804	\$28,158	312,391

Age At First Credit Report

A. Without ZIP5 F.E.

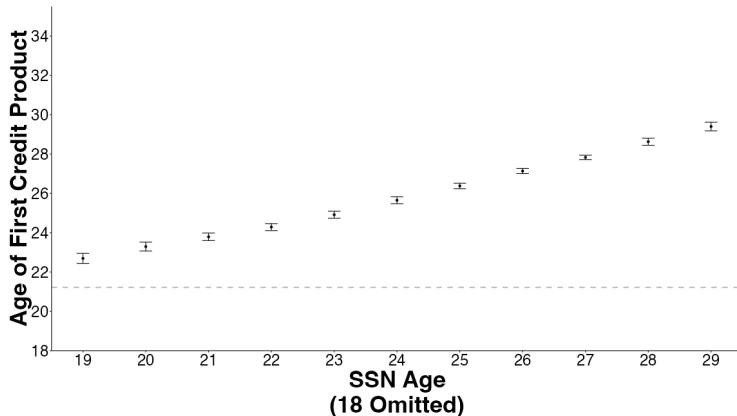


B. With ZIP5 F.E.



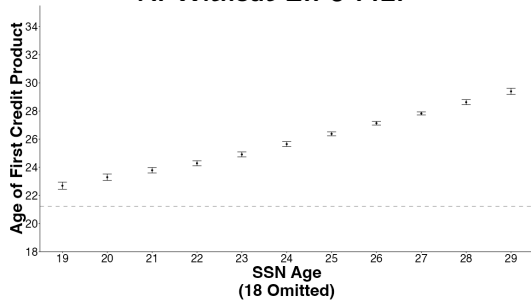
SSN Assigned At Older Age $\rightarrow$ Later Entry Into U.S. Credit System

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$

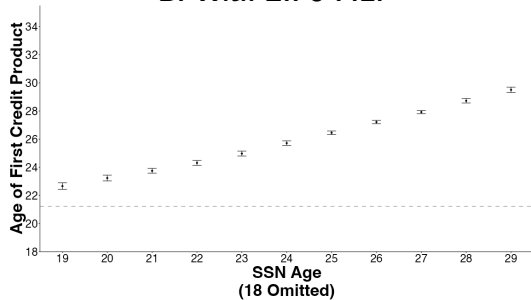


Age At First Credit Product

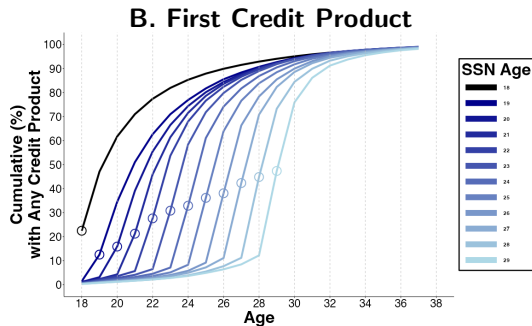
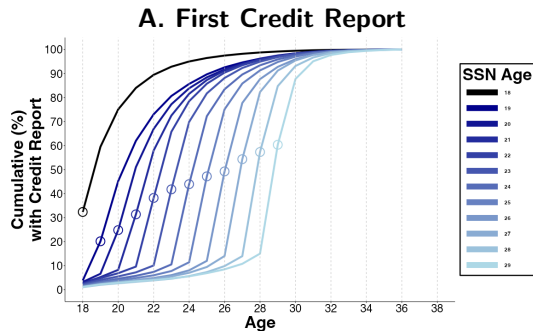
A. Without ZIP5 F.E.



B. With ZIP5 F.E.



Age At First Credit Report and First Credit Product By SSN Age Cohort



Credit Entry

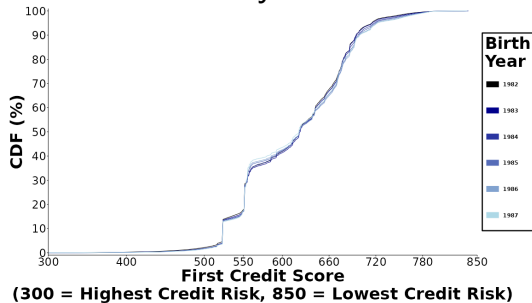
Dep Var: Age at First...	Credit Report		Credit Product	
	(1)	(2)	(3)	(4)
Immigrant	4.979*** (0.143)	1.808*** (0.088)	4.686*** (0.149)	1.618*** (0.106)
SSN Age		0.744*** (0.014)		0.720*** (0.016)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.264	0.289	0.120	0.129
N	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	19.864	19.864	21.251	21.251

Credit Entry With Additional F.E.

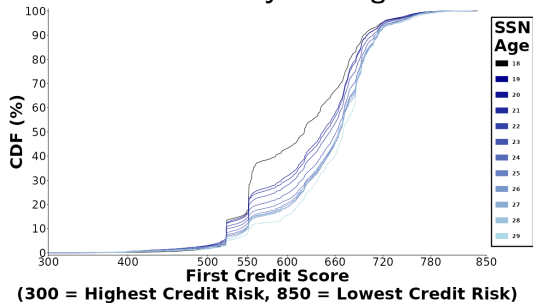
Dep Var: Age at First...	<u>Credit Report</u>		<u>Credit Product</u>		<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	4.782*** (0.117)	1.691*** (0.073)	4.403*** (0.116)	1.477*** (0.088)	3.719*** (0.206)	1.424*** (0.136)	1.642*** (0.101)	-0.082 (0.052)	3.929*** (0.146)	1.195*** (0.125)
SSN Age		0.727*** (0.014)		0.689*** (0.016)		0.540*** (0.023)		0.406*** (0.017)		0.644*** (0.019)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.299	0.323	0.184	0.192	0.168	0.169	0.153	0.154	0.155	0.159
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	19.864	19.864	21.251	21.251	29.151	29.151	36.356	36.356	22.677	22.677

CDF Of First Credit Score 🚀

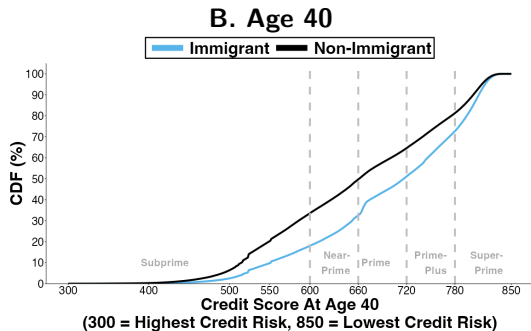
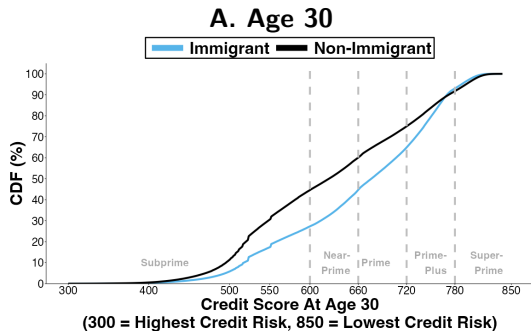
A. CDF By Birth Year



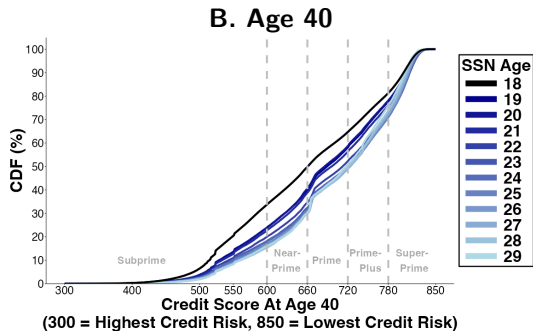
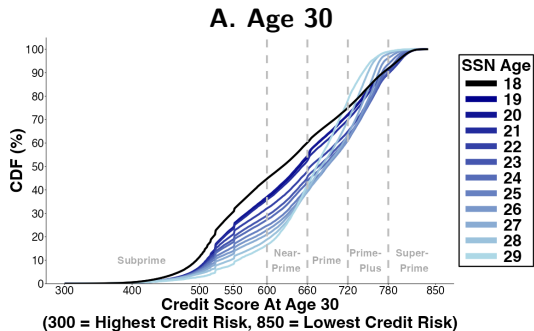
B. CDF By SSN Age



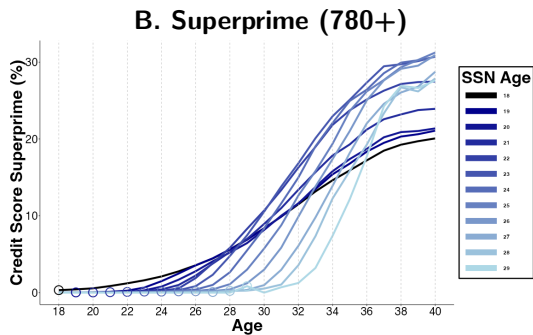
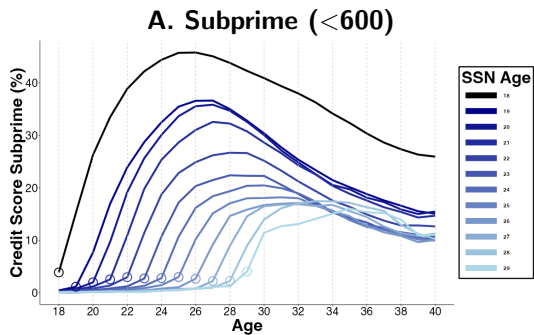
CDF Of Credit Score At Ages 30 & 40, By Immigration Status 🗺️



CDF Of Credit Score At Ages 30 & 40, By SSN Age 🗺️



Lifecycle Of The Distribution Of Credit Scores By SSN Age Cohorts 🚀



Immigrant Credit Scores Higher In Spite Of Shorter Credit Histories

	Credit Score at		Any Prime or Higher (660+) Credit Score at	
	Age 30	Age 40	Age 30	Age 40
	(1)	(2)	(3)	(4)
Immigrant	15.7*** (1.2)	21.6*** (0.6)	6.6*** (0.6)	2.0*** (0.2)
SSN Age	1.6*** (0.3)	1.5*** (0.2)	-0.1 (0.2)	1.6*** (0.1)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.202	0.200	0.156	0.152
N	5,755,134	4,449,929	6,122,932	4,800,195
Mean, SSN Age <21	626.1	659.5	37.71	46.84

Credit Scores At Ages 30 & 40, 🇵🇸

Dep Var: Credit Score at ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	27.1*** (1.1)	18.8*** (1.6)	32.7*** (1.0)	25.8*** (0.7)
SSN Age		2.0*** (0.4)		1.5*** (0.2)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.135	0.135	0.132	0.132
N	5,755,134	5,755,134	4,449,929	4,449,929
Mean, SSN Age j21	626.1	626.1	659.5	659.5

Credit Scores At Ages 30 & 40, With Additional F.E. 📌

	Credit Score At...				Probability of Prime or Higher Score At...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	22.27*** (1.03)	15.69*** (1.24)	28.14*** (0.94)	21.56*** (0.64)	5.99*** (0.92)	6.60*** (0.57)	8.97*** (0.27)	1.99*** (0.21)
SSN Age		1.61*** (0.31)		1.46*** (0.16)		-0.14 (0.19)		1.58*** (0.09)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.202	0.202	0.200	0.200	0.156	0.156	0.151	0.152
N	5,755,134	5,755,134	4,449,929	4,449,929	6,122,932	6,122,932	4,800,195	4,800,195
Mean, SSN Age j21	626.10	626.10	659.52	659.52	37.71	37.71	46.84	46.84

Credit Score Prime or Higher (660+) At Ages 30 & 40, 🍷

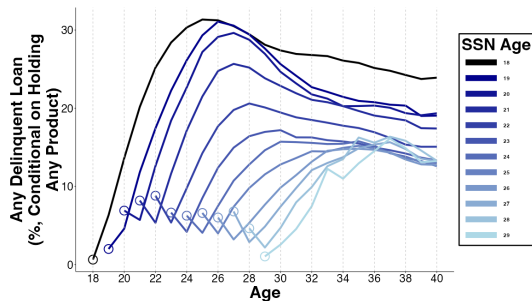
Dep Var: Prime or Higher at ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	7.30*** (1.02)	7.63*** (0.74)	9.00*** (0.20)	2.01*** (0.14)
SSN Age		-0.08 (0.20)		1.58*** (0.09)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.104	0.104	0.092	0.092
N	6,122,932	6,122,932	4,800,195	4,800,195
Mean, SSN Age j21	37.71	37.71	46.84	46.84

Credit Score Prime or Higher (660+) At Ages 30 & 40, With Additional F.E.



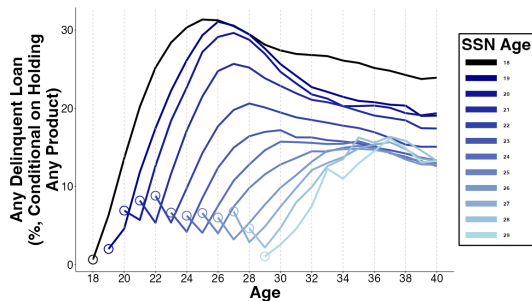
Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency

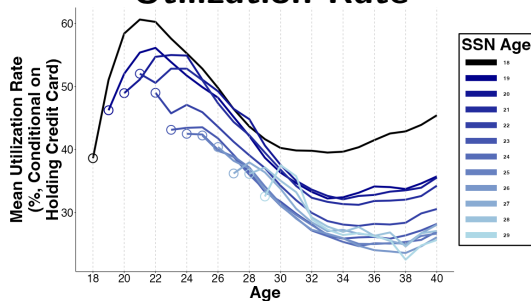


Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency



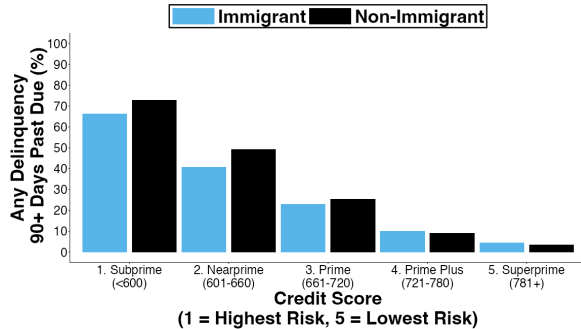
B. Credit Card Utilization Rate



↓ 5% delinquency rates at age 37 conditional on credit score at age 30

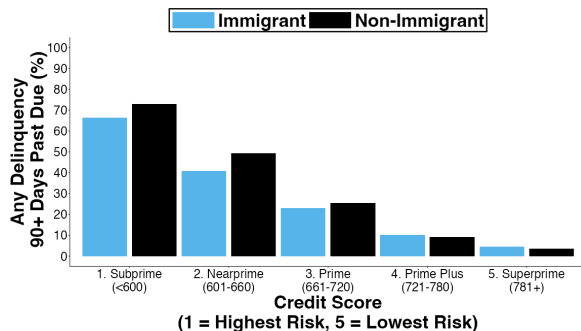
Immigrants (vs. Non-Immigrants) Are More Creditworthy:

5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



Immigrants (vs. Non-Immigrants) Are More Creditworthy:

5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



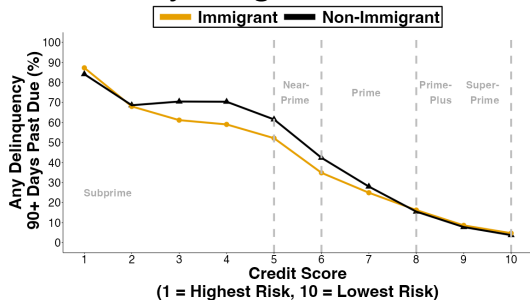
Immigrant	-0.0140*** (0.0041)
SSN Age	-0.0004 (0.0005)
F.E. Birth Year	X
F.E. Age 30 Zip5	X
F.E. Age 30 Credit Score	X
R^2	0.185
N	5,755,134
Mean, SSN Age <21	0.2722

Calibration Bias: Delinquency At 37 Conditional On Age 30 Credit Score

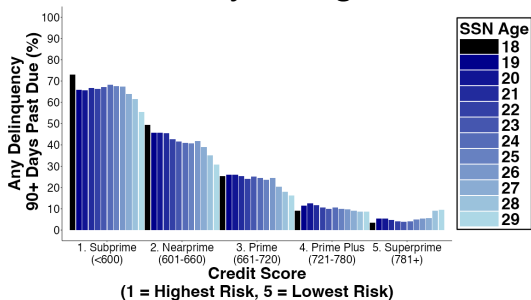
Dep Var: Any delinquency by age 37	<u>All</u>		<u>Subprime</u>		<u>Nearprime</u>		<u>Prime</u>		<u>Prime Plus</u>		<u>Superprime</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-1.58*** (0.24)	-1.40*** (0.41)	-2.85*** (0.40)	-3.69*** (0.74)	-3.53*** (0.36)	-2.72*** (0.58)	-1.49*** (0.24)	0.06 (0.49)	0.08 (0.10)	0.99*** (0.19)	0.28 (0.14)	0.37 (0.28)
SSN Age		-0.04 (0.05)		0.23 (0.12)		-0.19 (0.09)		-0.34*** (0.07)		-0.21*** (0.03)		-0.03 (0.07)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X	X	X
R^2	0.185	0.185	0.032	0.032	0.028	0.028	0.027	0.027	0.025	0.025	0.022	0.022
N	5,755,134	5,755,134	2,515,933	2,515,933	898,118	898,118	871,578	871,578	999,130	999,130	999,130	470,375
Mean, SSN Age j21	27.22	27.22	45.53	45.53	26.20	26.20	13.14	13.14	4.51	4.51	1.63	1.63

Any Delinquency By Age 37, Conditional On Age 30 Credit Score 📍

A. By Immigration Status



B. By SSN Age

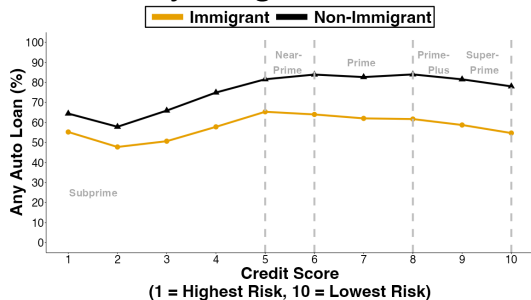


Age 37 Credit Outcomes Conditional On Age 30 Credit Score 🍷

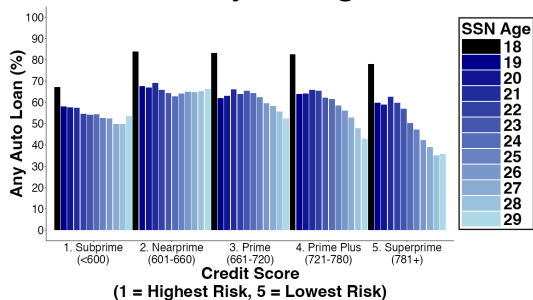
Dep Var: Outcomes at Age 37...	<u>Any Auto Loan</u>		<u>Any Mortgage</u>		<u>Any Credit Card</u>		<u>Number of Credit Cards</u>		<u>Credit Card Limits</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-14.68*** (0.49)	-8.89*** (0.39)	-1.42*** (0.06)	-1.61*** (0.13)	-11.93*** (0.49)	-2.33*** (0.50)	0.40*** (0.04)	0.22*** (0.03)	-795.8 (565.0)	981.1 (561.8)
SSN Age		-1.42*** (0.06)		0.05 (0.03)		-2.35*** (0.12)		0.04*** (0.01)		-434.6** (131.7)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X
R^2	0.107	0.108	0.268	0.268	0.110	0.110	0.142	0.142	0.306	0.306
N	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134	5,755,134
Mean, SSN Age j21	75.34	75.34	44.73	44.73	92.21	92.21	3.92	3.92	\$20,515.6	\$20,515.6

Any Auto Loan By Age 37, Conditional On Age 30 Credit Score 🚀

A. By Immigration Status

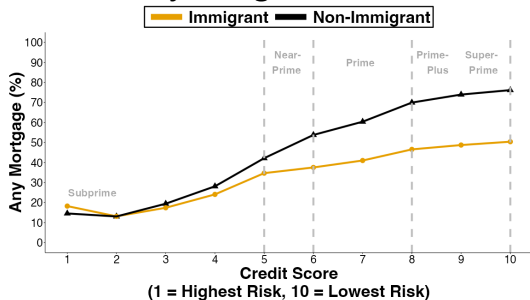


B. By SSN Age

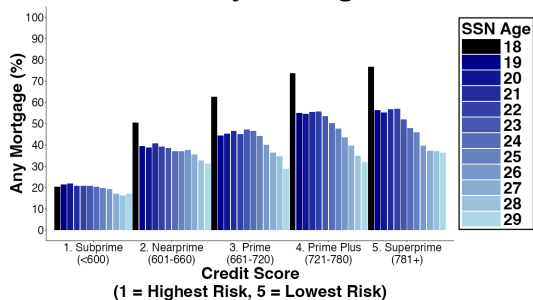


Any Mortgage By Age 37, Conditional On Age 30 Credit Score 🚀

A. By Immigration Status

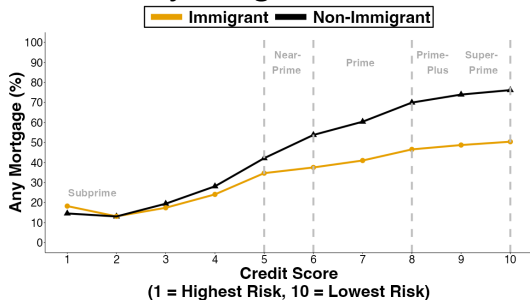


B. By SSN Age

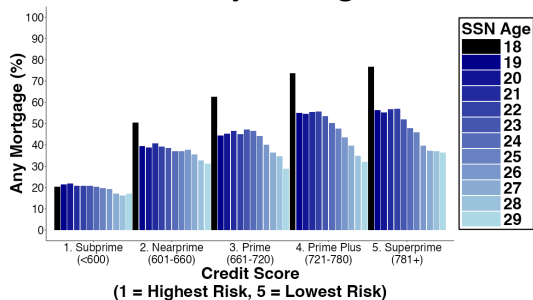


Any Mortgage By Age 37, Conditional On Age 30 Credit Score 🚀

A. By Immigration Status

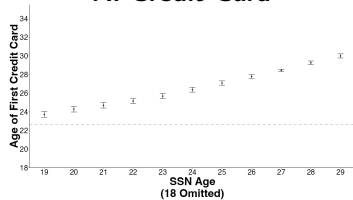


B. By SSN Age

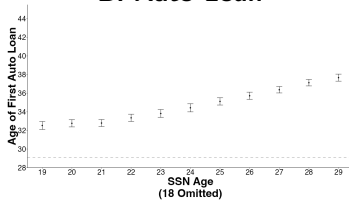


Age At First Credit Product By Type 🚀

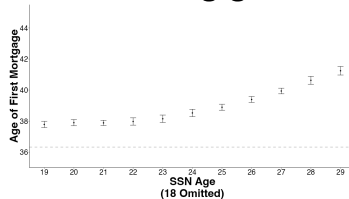
A. Credit Card



B. Auto Loan



C. Mortgage



Dep Var: Age at First...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	4.604*** (0.274)	1.967*** (0.217)	2.058*** (0.151)	0.219* (0.083)	4.309*** (0.184)	1.367*** (0.158)
SSN Age		0.619*** (0.024)		0.432*** (0.018)		0.690*** (0.020)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.080	0.082	0.080	0.081	0.080	0.085
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	29.151	29.151	36.356	36.356	22.677	22.677

Credit Access With Additional F.E.

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	4.782*** (0.117)	1.691*** (0.073)	4.403*** (0.116)	1.477*** (0.088)	3.719*** (0.206)	1.424*** (0.136)	1.642*** (0.101)	-0.082 (0.052)	3.929*** (0.146)	1.195*** (0.125)
SSN Age		0.727*** (0.014)		0.689*** (0.016)		0.540*** (0.023)		0.406*** (0.017)		0.644*** (0.019)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.299	0.323	0.184	0.192	0.168	0.169	0.153	0.154	0.155	0.159
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	19.864	19.864	21.251	21.251	29.151	29.151	36.356	36.356	22.677	22.677

Credit Access By Age 37

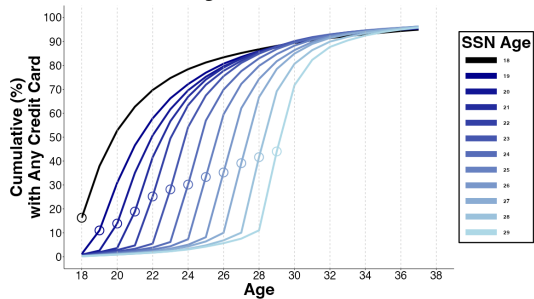
Dep. Var: By Age 37, has ...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-12.99*** (0.80)	-6.87*** (0.46)	-7.18*** (0.74)	0.39 (0.53)	0.31 (0.22)	0.15 (0.16)
SSN Age		-1.44*** (0.10)		-1.77*** (0.09)		0.04 (0.03)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.039	0.039	0.062	0.063	0.025	0.025
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	77.22	77.22	45.46	45.46	94.93	94.93

Credit Access By Age 37 With Additional F.E.

Dep Var: By Age 37, has...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-8.94*** (0.67)	-4.43*** (0.20)	-4.30*** (0.67)	2.33*** (0.35)	1.14*** (0.22)	0.50*** (0.14)
SSN Age		-1.06*** (0.11)		-1.56*** (0.09)		0.15*** (0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.124	0.124	0.146	0.146	0.060	0.060
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age $\bar{x}_{21}$	77.22	77.22	45.46	45.46	94.93	94.93

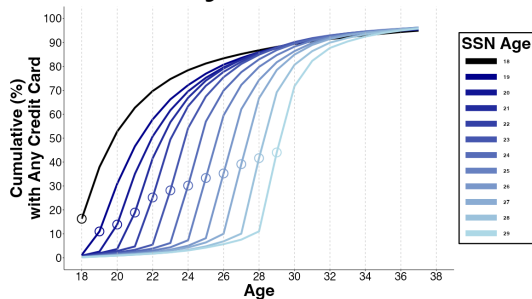
Immigrants' Access to Credit Cards Converges With Non-Immigrants

A. Any Credit Card

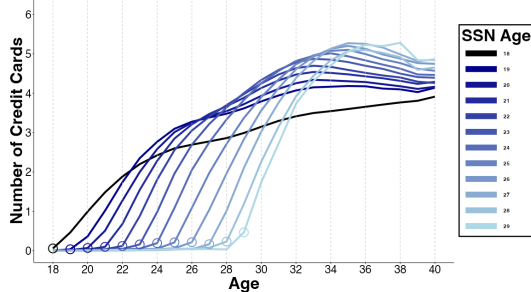


Immigrants' Access to Credit Cards Converges With Non-Immigrants

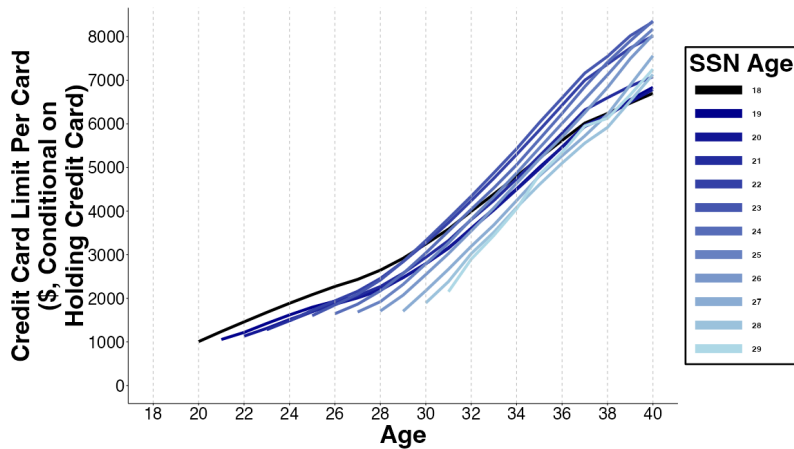
A. Any Credit Card



B. Number of Credit Cards

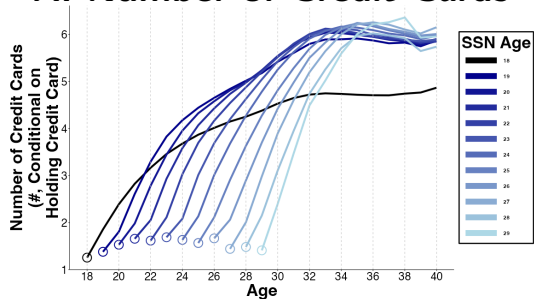


Credit Card Limit Per Card 🚀



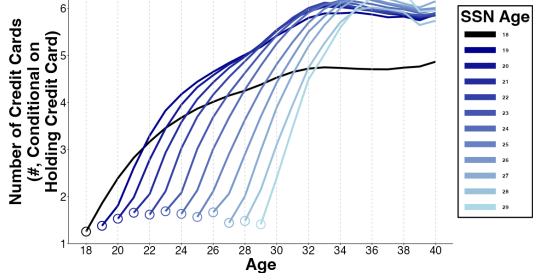
Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

A. Number of Credit Cards

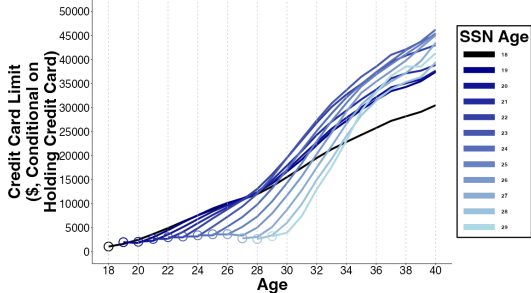


Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

A. Number of Credit Cards



B. Credit Card Limits



Number of Credit Cards At Age 30 🚀

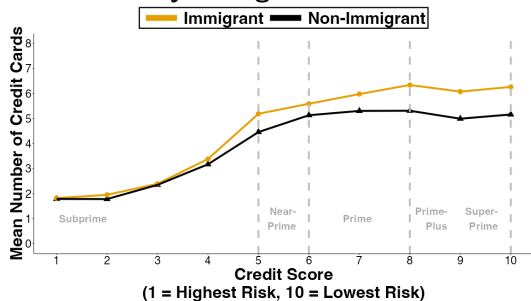
Number of Credit Cards At Age 40 🚀

Credit Card Limits At Age 30 🚀

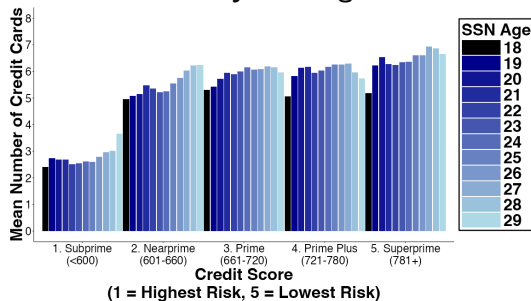
Credit Card Limits At Age 40 🚀

Number Of Credit Cards At Age 37, Conditional On Age 30 Credit Score 📍

A. By Immigration Status

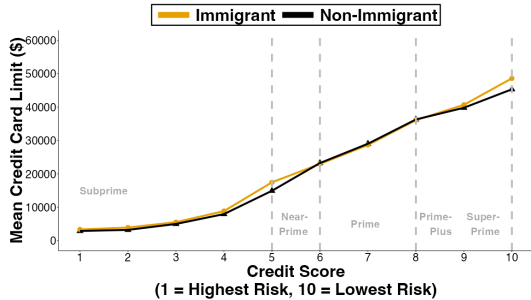


B. By SSN Age

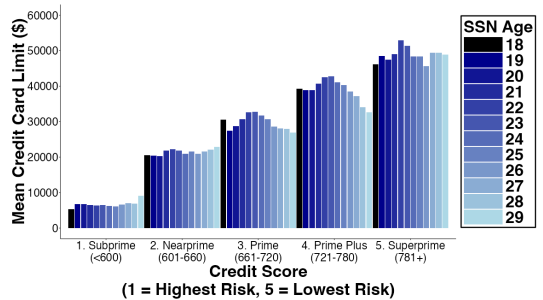


Credit Card Limits At Age 37, Conditional On Age 30 Credit Score 🚀

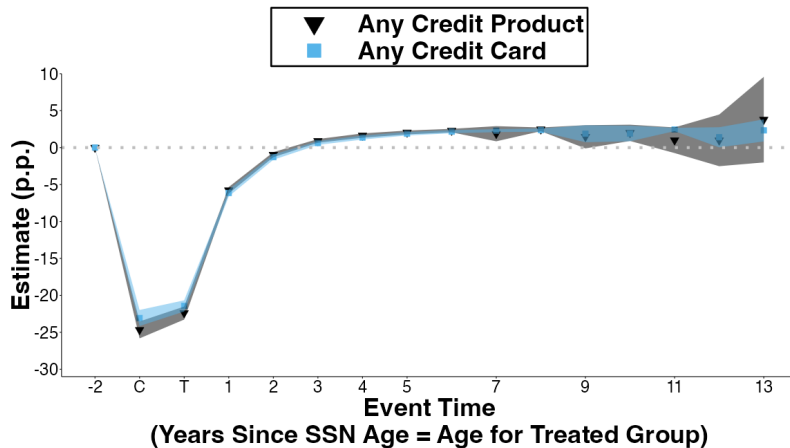
A. By Immigration Status



B. By SSN Age

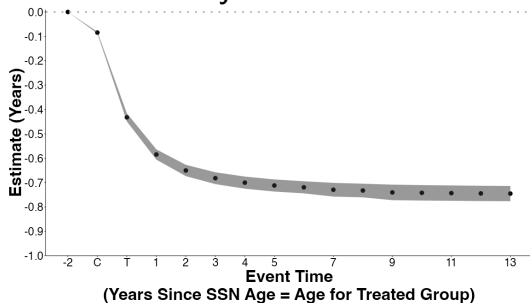


Paired Cohorts: First Credit Product & First Credit Card 🚀

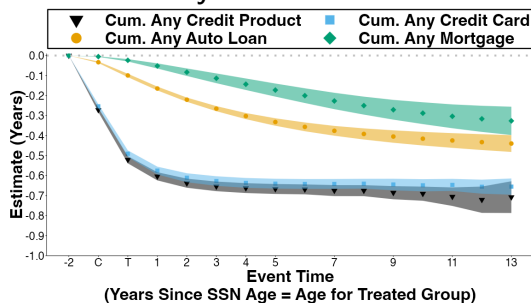


Paired Cohorts: Cumulatives 🚀

A. Any Credit Score



B. By Credit Product

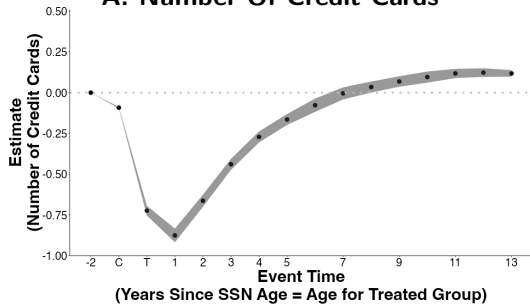


Paired Cohorts: Cumulative Years of Delay 🚀

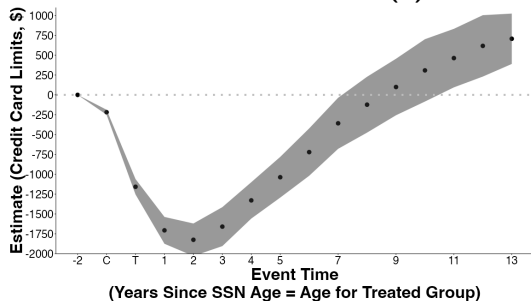
Event Time	<u>Any Credit Score</u> (1)	<u>Credit Product</u> (2)	<u>Auto Loan</u> (3)	<u>Mortgage</u> (4)	<u>Credit Card</u> (5)
C	-0.085*** (0.002)	-0.272*** (0.005)	-0.033*** (0.002)	-0.005*** (0.001)	-0.253*** (0.005)
T	-0.432*** (0.009)	-0.521*** (0.009)	-0.099*** (0.004)	-0.024*** (0.003)	-0.491*** (0.008)
T+5	-0.712*** (0.013)	-0.667*** (0.013)	-0.332*** (0.01)	-0.173*** (0.017)	-0.640*** (0.013)
T+10	-0.742*** (0.016)	-0.690*** (0.018)	-0.415*** (0.017)	-0.288*** (0.028)	-0.648*** (0.016)
<i>N</i> Paired Cohorts	68	68	68	68	68
<i>N</i> Consumers	403,506	403,506	403,506	403,506	403,506
<i>N</i>	6,456,096	6,456,096	6,456,096	6,456,096	6,456,096

Paired Cohorts: Credit Cards 🚀

A. Number Of Credit Cards



B. Credit Card Limits (\$)



Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

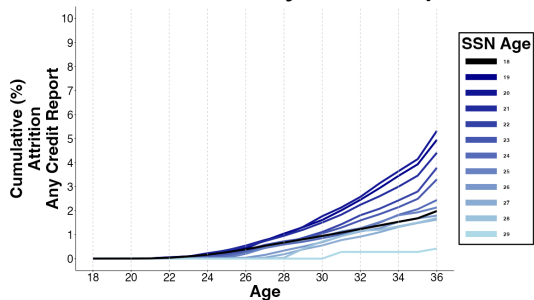
Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

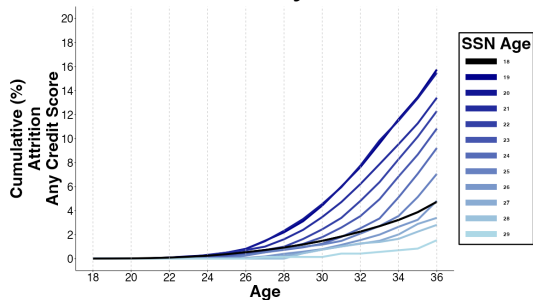
Restrict to consumers observed in 2024:

- **Mortgage** Immigrant vs. Non-Immigrant gap closes but SSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

A. Attrition In Any Credit Report

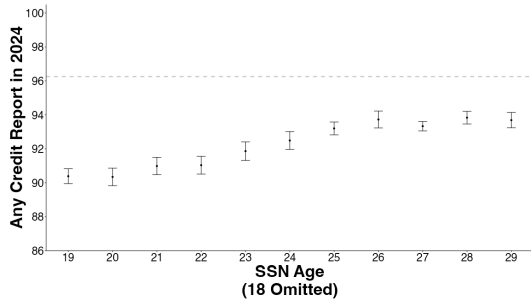


B. Attrition In Any Credit Score

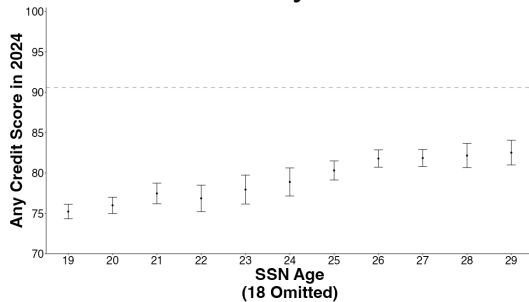


Consumers Observed In 2024 🚀

A. Attrition In Any Credit Report

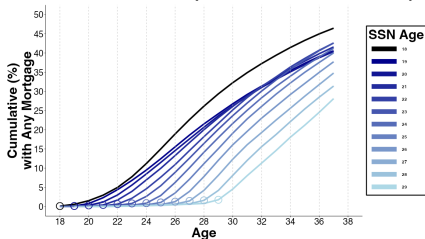


B. Attrition In Any Credit Score

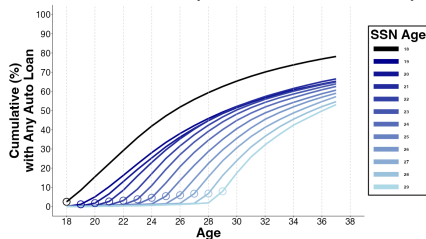


Lifecycle Of Credit For Consumers Observed In 2024 📍

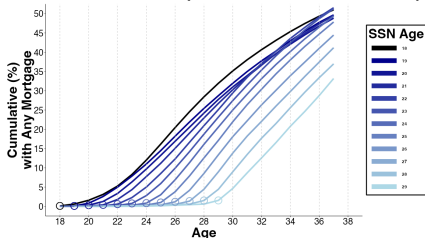
A. Mortgages (Any 2024 Report)



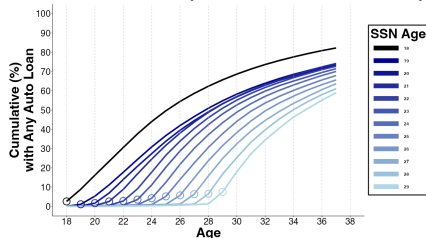
B. Auto Loan (Any 2024 Report)



C. Mortgages (Any 2024 Product)

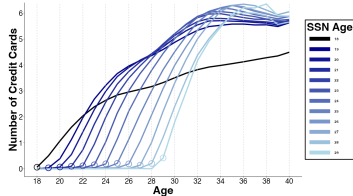


D. Auto Loan (Any 2024 Product)

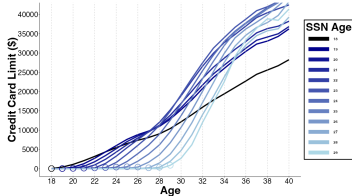


Lifecycle Of Credit For Consumers Observed In 2024 (Any Tradeline) 🚀

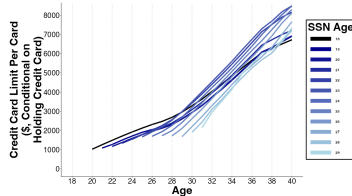
A. Number Credit Cards



B. Credit Card Limits



C. Credit Limit Per Card



Consumers Observed In Data In 2024 With Additional F.E. 📍

Dep Var: In 2024, has...	Any Credit Report		Any Credit Score		Any Credit Tradeline	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-2.64*** (0.13)	-4.85*** (0.22)	-7.52*** (0.29)	-12.16*** (0.48)	-7.50*** (0.32)	-12.07*** (0.42)
SSN Age		0.52*** (0.04)		1.09*** (0.07)		1.08*** (0.06)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.034	0.034	0.110	0.111	0.138	0.138
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age j21	96.15	96.15	90.31	90.31	85.03	85.03

Credit Access By Age 37 With Additional F.E. Conditional On Any Credit Score Observed in 2024 📍

Dep Var: By Age 37, has...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-8.30*** (0.81)	-2.01*** (0.22)	-2.89*** (0.71)	6.57*** (0.41)	1.03*** (0.20)	0.67*** (0.13)
SSN Age		-1.46*** (0.12)		-2.20*** (0.11)		0.08 (0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.099	0.099	0.134	0.134	0.054	0.054
N	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726
Mean, SSN Age $\downarrow$ 21	80.37	80.37	48.76	48.76	96.05	96.05

Use Survey of Consumer Finances (SCF) 2022 To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{Immigrant}\}_i + \beta_2 \cdot \text{Immigration Age}_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$ are F.E. Demographic Controls: male, race, hispanic, spouse, male spouse, marital status, household size, number of children, education levels, & quintiles of household wage income

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SCF Debt Attitude Questions:

“People have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...to finance the purchase of a car.”

Immigrants More Likely To Buy Cars With Cash (SCF, 2022)

12.5 p.p. [95% C.I. -2.2, 27.3], 17% Vs. Non-Immigrants

Dep Var:	<u>Has Car</u>		<u>Any Auto Debt</u>		<u>Homeowner</u>		<u>Any Mortgage</u>		<u>Credit Card Limits</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-7.13*	-6.14	-4.27	-11.14	-13.55***	0.35	-13.09***	-5.27	-3,078.8	6,373.8
	(3.15)	(4.96)	(5.08)	(8.27)	(4.64)	(6.85)	(4.64)	(7.22)	(3,460.2)	(6,316.0)
Immigration Age		-0.13		0.88		-1.78***		-1.00		-1,213.9*
		(0.47)		(0.76)		(0.59)		(0.64)		(582.4)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.110	0.110	0.063	0.063	0.258	0.261	0.258	0.259	0.254	0.256
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age j21	89.23	89.23	47.14	47.14	53.98	53.98	45.25	45.25	\$20,275.9	\$20,275.9

*** 1%, **, 5%, * 10% statistical significance

Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

Dep Var: Apply For...	<u>Any Credit</u>		<u>Auto</u>		<u>Mortgage</u>		<u>Credit Card</u>		<u>Other Credit</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32 (4.56)	-1.08 (7.17)	2.03 (4.37)	-6.55 (7.53)	2.11 (3.8)	5.65 (6.85)	1.63 (3.22)	2.02 (5.32)	1.84 (3.62)	-7.66 ⁺ (4.61)
Immigration Age		0.44 (0.62)		1.10 (0.72)		-0.45 (0.65)		-0.05 (0.48)		1.22* (0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age ≥ 21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08

*** 1%, ** , 5%, * 10% statistical significance

Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

Dep Var: Apply For...	<u>Any Credit</u>		<u>Auto</u>		<u>Mortgage</u>		<u>Credit Card</u>		<u>Other Credit</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32 (4.56)	-1.08 (7.17)	2.03 (4.37)	-6.55 (7.53)	2.11 (3.8)	5.65 (6.85)	1.63 (3.22)	2.02 (5.32)	1.84 (3.62)	-7.66 ⁺ (4.61)
Immigration Age		0.44 (0.62)		1.10 (0.72)		-0.45 (0.65)		-0.05 (0.48)		1.22* (0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age ≥ 21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08

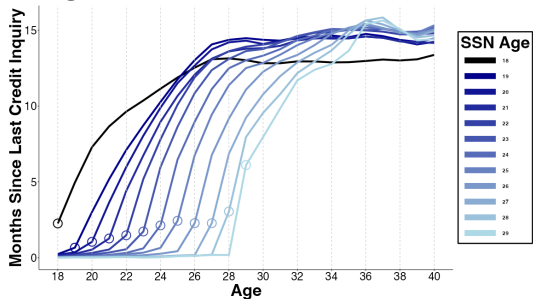
*** 1%, ** , 5%, * 10% statistical significance

Credit inquiries data in credit panel shows
↑ immigrant credit demand in 20s, ↓ demand in 30s

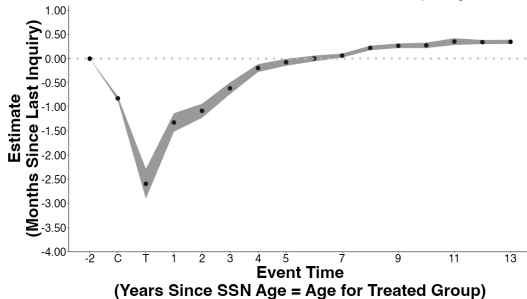
Does Lower Credit Demand Explain Lower Access?

Immigrants Have $\uparrow$ Credit Demand In 20s, $\downarrow$ Demand In 30s 🚀

A. Months Since Last Credit Inquiry
Higher Value = Lower Credit Demand



B. Paired Cohorts
Months Since Last Credit Inquiry



Caveat: Lower credit demand in 30s may be function of earlier rejections $\rightarrow$ deterrence

Inconclusive Evidence On Other Demand and Supply Proxies (SCF, 2022)

...Given Low Statistical Power To Detect Out Differences

(Conditional On... Dep Var:	Why Did Not Apply For Credit?				Outcome After Applying For Credit?					
	...Not Applying)				...Applying)		...Reject)		...Reject & Reapply)	
	No Demand		Rate Too High		Reject/Less		Reapply		Reapply Reject	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	0.41 (7.98)	6.84 (12.64)	8.01 (5.85)	-0.73 (8.25)	-4.95 (4.74)	-2.28 (7.43)	3.43 (14.11)	-10.42 (22.95)	46.10*** (14.27)	49.69** (18.81)
Immigration Age		-0.88 (1.03)		1.20 (0.87)		-0.33 (0.61)		1.73 (1.89)		-0.40 (1.68)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.155	0.155	0.090	0.092	0.054	0.054	0.063	0.062	0.312	0.307
N	666	666	666	666	1,147	1,147	292	292	178	178
Mean, Immigration Age ≥ 21	79.52	79.52	7.53	7.53	27.07	27.07	60.30	60.30	60.22	60.22

*** 1%, **, 5%, * 10% statistical significance